Precompliance Testing Using Software Defined Radio

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***Abstract*— According to the increasing demand for EMC (Electromagnetic Compatibility) measurements, EMC labs and their equipment might work on maximum capacity. On the other hand, in earlier stages of hardware development, full compliance testing is not necessarily needed. This gives us a clue that a cheaper alternative for measurements should be searched, but not on all costs. There is some basic equipment which should be purchased or rented, but secondary or researching equipment might be cheaper than some cheaper alternatives. In this article we search how a fully compliant test receiver may be replaced with cheaper alternatives, for example using an SDR (Software Defined Radio) instead of a real test receiver. What are the capabilities or boundaries of them, but taken into consideration that proper equipment is always necessary for proper results. As a beginning some parameters of the used "receivers" should be measured with network analysers before connecting other equipment to the input and making some major damage. Next to SDRs, other tools can be taken into consideration for example pre-compliance testers, oscilloscopes, or spectrum analysers.**

Keywords—: EMC; measuring receiver; SDR; spectrum analyser; oscilloscope; EMC test equipment

I. Introduction

In all department of electronic design and manufacturing, electromagnetic compatibility (EMC) is one of the most important factors. All the devices need electromagnetic compatibility to work properly. Some of them can be interfere with others easily, while others radiate lots of electromagnetic waves. Nowadays EMC is becoming more important than ever, because of the growing number of electronic devices, gadgets, smart devices we use.

The continuous development of technology causes Industry 4.0 become the description of our daily life. As the size of electronic circuits and their manufacturing prices constantly reducing besides the fact their calculational power increasing, we can make smarter and smarter devices and systems. These smart systems and devices allow more precise, optimized, and accurate control of different household devices, industrial production lines and even military devices. This development has (if has) one negative side effect in EMC.

II. The purpose of this paper

The increasing need of EMC testing generates the need for developers to search EMC parameters in earlier stages without serious costs. EMC labs do research and development measurements, but these might be costly. Some of the parameters can be checked without proper EMC lab equipment and this topic is also about a cost-effective solution. Quality laboratory tools cannot be avoided, but as secondary equipment, or for true researching measurements some cheaper equipment can be used. One of the most expensive EMC laboratory equipment is the measuring receiver. Depending on the needs it has a wide range of frequency measuring capabilities, measuring technologies and precise parameters. On the other hand, these receivers are high quality spectrum analysers produced specially to fulfil the EMC measurement standards. For a very different and so much cheaper tool, we can consider SDRs. These tiny (not always so tiny) tools can also act as spectrum analysers, but with some restrictions. At first, here is a brief introduction about this two equipment.

III. Different receivers

A. TRF (Tuned Radio Frequency)

A receiver is a circuit or system which capture a given bandwidth of signals in the EM spectrum. Most receivers focus only on one or maybe a few signals, but not only receive them, but also recovers the needed decoded signal which modulated the carrier. It is commonly used as a communication link. The simplest radio receiver is the tuned radio frequency which is basic bandpass filter (selecting the desired signal). These tuned radio frequency design can be built for receiving frequencies form tens of kHz to many MHz. As an example, the classic crystal radio is a perfect example for the tuned radio frequency. [1]

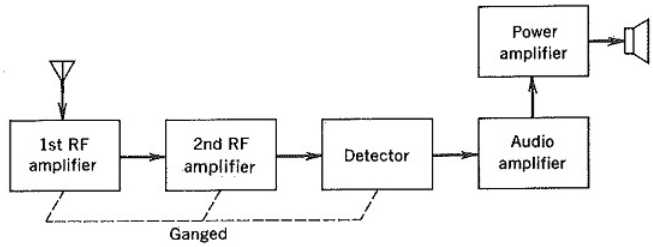


Fig. 1. The TRF receiver [2]

B. Superheterodyne receiver

It can be clearly seen that simple solutions work in specialized and limited cases (but for these purposes they are perfect and simple). The next stage of receivers is the superheterodyne approach. It uses positive feedback in the tuned radio frequency design to increase gain. The variable tuning is achieved via a local oscillator and filtering. The superheterodyne receiver can be used with any type of

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modulation. The key factors of its success are the stability, repeatability, and purity of the local oscillator. This oscillator was used to be built as an analogue oscillator using vacuum tubes, transistors, and ICs. But nowadays it is replaced with digitally synthesized oscillators. The superheterodyne receivers can achieve a high frequency range, but for accuracy and stability reasons, double conversions and even triple conversions are used to work optimized. [1]

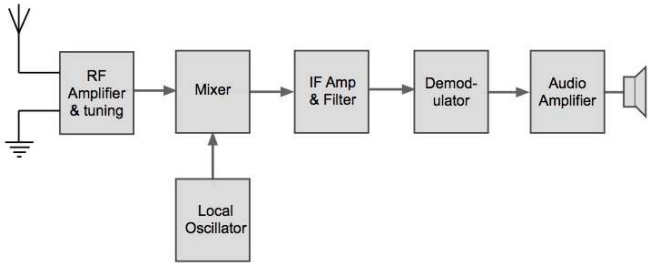


Fig. 2. Superheterodyne receiver [3]

C. Zero-IF receiver

The next type of receivers is the zero-IF receiver design. It is also called a direct conversion or homodyne architecture which uses a local oscillator tuned to the desired carrier frequency or close to it. In this case the IF frequency is zero or near zero. With this solution IF-related problems are avoided (for example filtering) and very wideband RF signals can also be demodulated. The disadvantage of the zero-IF is that it has extreme demands on the LO and the mixer performance. Also dynamic range problems are come with this type. [4]

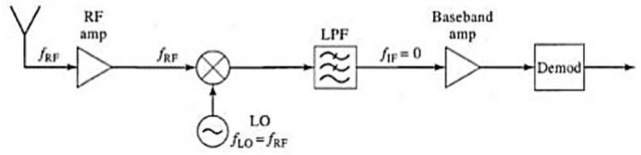


Fig. 3. Zero-IF receiver [5]

IV. SDR (Software Defined Radio)

The SDR is a different approach to tuning, demodulating, and decoding a received signal. It does not contain analogue circuitry and IF stages and not even uses the zero-IF approach (but it works like a zero-IF receiver). It uses fast, precise, and high-resolution ADC (analogue-digital conversion) of the received signal at its original carrier frequency. The data stream uses complex algorithms in software to find the desired signal, extract it and recover the modulating data. [6]

The biggest advantage of the SDRs is the flexibility. The same circuitry can recover signals in a wide band of the spectrum and with different bandwidths and modulation processes. It is not limited to specific signals and can be changed on the fly. It is very attractive in military and public safety situations because it does not have any interoperability issues. The design can handle AM, FM, PM modulations can recover data from different formats because all of it is done by software.

The weakness of the SDR is at first the ADC circuit. It must be a typically 12- or 14-bits conversion resolution AD converter at high speed and wide bandwidth. Any problem caused by the ADC and the data can be corrupted and it misleads the algorithms lead to further data errors or require extra processing. The data stream usually analysed by FPGAs and none of these components are the cheapest in electronic technologies. [7]

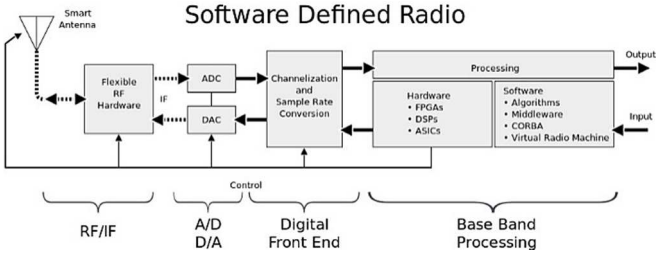


Fig. 4. Software Defined Radio [8]

V. SDR instead of measuring receiver

The concept of EMC measurements is based around a measuring receiver (about half of the standard measurements). But it is an expensive instrument. It has a wide frequency range, contain different detectors (peak, average, quasi-peak, RMS, and others), impedance matched input ports with a wide dynamic range but great sensitivity. So, the question about using a tool which does not have any of these features is valid. Of course a cheap SDR cannot replace or not even get closer in performance to measuring receivers, but the price of it, and the receiving capabilities can give us a feeling of why not to try it. [9]

A. ESR26 Measuring receiver

If we want to purchase a decent measuring receiver, we easily find the Rohde & Schwarz instruments. There are a couple of different receivers provided by this company but to be specific we talk about an ESR-26 type, because it will be used later for the measurements. A measuring receiver is a triple stage superheterodyne receiver. This receiver has a frequency range from 10 Hz to 26,5 GHz, has an integrated 20 dB preamplifier. It uses RBWs (Resolution Bandwidth) from 10Hz to 1 MHz, while the real time spectrum analysis has a span up to 40 MHz. [10]

It has an ultrafast time domain scan or the conventional stepped frequency scan. It communicates with computers using GPIB or LAN but can be used standalone. Most of these features are used in everyday EMC measurements. On the other hand, if we think about engineering or research measurements, not all of these are required. If we know the problematic emission frequencies of the product, further development may require only simple devices for measurements such as spectrum analysers, oscilloscopes or SDRs. [11]



Fig. 5. R&S ESR26 measuring receiver

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B. *RTL-SDR Software Define Radio*

The SDR used by us is an RTL-SDR with the R820T chip inside. It is the cheapest SDR on the market, but considering the capabilities of the other available SDRs, it is worth its price.



Fig. 6. RTL-SDR Software Defined Radio [12]

TABLE I. AVAILABLE SDRS ON THE MARKET [13]

SDR	Tune Low (MHz)	Tune Max (MHz)	RX BW (MHz)	ADC Res. (Bits)	Transmit? (Yes/No)	Price ($USD)
RTL-SDR (R820T)	24	1766	3.2 / 2.56	8	No	~20
Funcube Pro+	0.15 410	260 2050	0.192	16	No	~200
Airspy	24	1800	10	12	No	199
SDRPlay	0.1	2000	8	12	No	149
HackRF	30	6000	20	8	Yes	299
BladeRF	300	3800	40	12	Yes	400 & 650
USRP 1	DC	6000	64	12	Yes	700

The R820T chip is a digital silicon tuner with low power consumption. It is designed for the digital TV standards and used in TVs, USB peripherals, set top boxes, media players and automobile TVs. It needs a digital demodulator to use with USB or PCIe. The RTL-SDR has an USB port on it. The R820T chip has a frequency range from 42 to 1002 MHz (it is not clear how the final SDR can be over it), has a 3.5 dB noise figure. The maximal input power is +10 dBm (which is important if we want to use it not only with antennas). The input return loss (S11) is -10 dB typically. [14]

VI. MEASURING THE VSWR OF THE SDR

As a start, we wanted to find out how the SDR can work with other equipment, so we checked the VSWR of it. Not directly, but it shows us how good is it not to return the signals into the source. For the measuring receiver it is typically lower than 2.2. The SDRs VSWR is from 1.3 to 3.7. The measurement setup consisted of a Rohde & Schwarz ZVH8. We didn't have a proper calibration kit for SMA connectors (only for N connectors), but the purpose was not to calibrate. We used an N-SMA converter to connect the RTL-SDR.

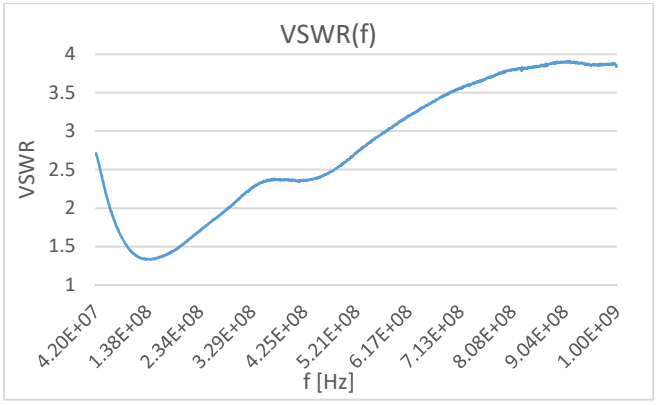


Fig. 7. The measured VSWR of the SDR

In a little range, the VSWR of the SDR is below 2.5. According to our opinion, based only on the VSWR values, the SDR can be used with restrictions.

For research purposes we measured the VSWR of a CEC (Conducted Emission Current) coil and a CEV (Conducted Emission Voltage) device. Both setups are using a similar range of frequency (like the usable range of the SDR), that is why we didn't measure any antennas or other equipment. This equipment usually needs preamps and for safety reasons some attenuator. Using preamps and attenuators, the VSWR of the SDR is not so important because it is radically changed by them.

VII. ACCURACY OF THE SDR

The next question about the SDR is the accuracy of it. In general, it does not need to be accurate, because its intended purpose is not to measure the different RF signals, but to receive them. A measuring receiver is a high precision device, but for research and development tests, it is not necessary.

As a next approach we measured the same RF source with both devices. The RF source was a RefRadX comb generator, its output was attenuated with 10 dB in order not to overdrive the input of the receiving devices. The biggest problem here was that the SDR does not work with resolution bandwidth, but with sampling rates. It is possible to calculate one into the other, but to compare the measured values we didn't do that. Here is a table which contains the measured data

TABLE II. MEASURING THE SAME RF SOURCE WITH BOTH DEVICES

f [MHz]	RTL-SDR [dBFS]	ESR26 [dBm]	Diff [dB]	U_{diff} [V]
10	-60	-14,68	55,32	583,4451
20	-56	-14,1	51,9	393,5501
50	-5,3	-16,7	-1,4	0,851138
100	-4,3	-16,7	-2,4	0,758578
200	-3,7	-15,6	-1,9	0,803526
500	-13,8	-22,7	1,1	1,135011
1000	-37,7	-37,2	10,5	3,349654

The table contains the measured value by the SDR in dBFS. This unit is a dB value based on the maximum input power. The maximum input power for the SDR is 10 dBm. The measuring receiver was set to show the values in dBm,

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so to compare the different values, 10 dB should be subtracted from the dBFS values. The column Diff shows the real difference between the measured values (in dBs), compensated with the 10 dBm. The U_{diff} shows the same value but calculated into voltage. The table clearly shows that the SDR can be used above 20 MHz, which is consistent to the datasheet (48 MHz). The first two rows are not real values in voltage. It also can be used till 1 GHz, but on that frequency, the difference increases.

These measurements can be improved later. For example, the VSWR according to the datasheet was measured with -10 dBm generator power, but we used -20 dBm on the network analyser. The accuracy measurement can be measured in more frequency points, and it may be determined how the different sampling rates effect the measured values. With our devices it is possible to measure not just the S11 value of the SDR, but also the transfer impedance of it. And as a final test, a compensation chart can be measured and using it, some real measurements could be done.

VIII. Conclusions

As a result, we can state that in special cases and with restrictions, the RTL-SDR can be used as a receiver in EMC measurements. At first, we introduced the basic concept of the different receiver types including the superheterodyne and zero-IF designs, because the used measuring receiver is a multiple stage heterodyne and the SDR is a zero-IF type. After we showed the most important parameters of them and provided data of two basic measurements. One of them was measuring the VSWR value of the SDR, the other was testing an RF source with both the SDR and the measuring receiver. According to the results, in a narrower frequency range the RTL SDR can be used in EMC testing. It is important to highlight that this type of SDRs is the cheapest one available on the market and its price cannot be compared to a calculated, professional measuring receiver.

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